

Vinesh Vangapandu

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Professional Summary

- Experienced data engineer with strong background in machine learning, data analysis, and software development. Skilled in multiple programming languages and data processing tools with focus on implementing efficient data solutions.
- Transforming raw data into strategic insights using Python, Hadoop, Hive, and Spark. My expertise spans building scalable ETL pipelines, designing MapReduce jobs for data optimization, and implementing machine learning models.
- Developed solutions such as AI-driven robotic automation using C++ as base, freight logistics optimization using the Pulp package in Python, and data-driven pricing strategies for retail liquor, increasing profitability by 15%.

Professional Experience

Axzora Resourcing | *ETL Developer with SAS* | *Manchester, NH.* Jul 2025 – Present

- Designing, developing, and maintaining ETL workflows using SAS Base, SAS EG, and SAS DI Studio to support data extraction, transformation, and loading processes.
- Analyzing complex data sources and transforming raw data into meaningful insights. Testing, and validating data transformation rules to ensure data consistency, accuracy, and reliability.

Welkin Technologies LLC | *Data Engineer* | *Alpharetta, GA.* Oct 2024 – Jun 2025

- Worked with the BI team to gather report requirements and Sqoop to export data to HDFS and Hive.
- Involved in data collection and treatment analytics by analyzing existing internal data and external data, worked on entry errors and classification errors, and defined criteria for missing values.
- Used cluster analysis for identifying customer segments, decision trees for profitable and non-profitable customers, and market basket analysis for customer purchasing behavior and part/product association.

Wright State University | *Research Assistant in AI* | *Fairborn, OH.* Jan 2024 – Apr 2024

- Worked with my Professor in developing a C++ code for simulation of a dual-robotic-arm system to rearrange blocks between two designated locations (L1 and L2).
- Designed robotic operations such as pickup, move, stack, unstack, and putdown to transform initial states into desired final configurations.
- Applied multithreading and optimized data structures to improve execution speed and system efficiency.

Nihar Info Global Limited | *Data Engineer Intern* | *Hyderabad, IND.* Jan 2021 - May 2021

- Developed multiple MapReduce jobs in Java for data cleaning and preprocessing; also assisted with data capacity planning and node forecasting. Also extracted and transformed the data sets, and the results were exported back to RDBMS using Sqoop.
- Utilizing advanced PL/SQL collections such as associative arrays, nested tables, and VARRAYs to handle complex data structures and improve the efficiency of data processing tasks.

Education

Wright State University | *MS in Computer Science* Aug 2021 – Apr 2024

- GPA: 3.2/4.0 (*Diploma-Certificate-link*)
- **Coursework:** Machine Learning, Design Algorithms and Analysis, Cloud Computing, Python Programming, Testing LLM models in Information Retrieval, Advanced Programming Language, Distributed Computing, React.

Gitam Deemed to be a University | *B-Tech in Computer Science* Aug 2017 – Apr 2021

- GPA: 7.08/10.00
- **Coursework:** Java Development, HTML, CSS, JavaScript, Introduction to AI, C++, MS Office, Probability and Statistics, Linux, Discrete Mathematics, Database Management Systems, Advanced Excel.

Projects

Robot Arms Block Simulation | C++, Multithreading, Data Structures, Terminal.

[visit-github](#)

- C++ simulation of a dual-robotic-arm system designed to rearrange blocks between two locations/warehouses (L1 and L2).
- It transforms an initial state of randomly arranged blocks into a user-defined final state by performing robotic arm operations such as pickup, move, stack, unstack, and putdown.
- This simulation models real-world logistics automation and manufacturing assembly line operations where robotic arms coordinate to manipulate objects.

Wine Sales Profit Optimizer Model using ML | Python, ML, Pandas, Scikit-learn

[visit-github](#)

- Atomic Wines, a Midwest retail chain, sought to identify underpriced wines in its wholesaler's catalog to maximize profit margins.
- Built a predictive model using wine chemical properties to estimate quality scores, enabling data-driven purchasing decisions for Atomic Wines by boosting profitability margins about 15%.
- Identifying the top 20 high-value wines by calculating the Predicted Quality/Price ratio, uncovering undervalued inventory with potential profit margin. Automated data preprocessing using Pandas, NumPy, and CSV output generation to streamline integration with business workflows.

Personal Web Portfolio with React-Vite | JavaScript, Tailwind CSS, React-Vite

[visit-github](#)

- Built a modern portfolio website with React and Vite.
- Showcases my projects, skills, and professional experience with smooth animations and responsive design using HTML, Tailwind CSS, and JavaScript.

Technologies

- **Programming Languages:** C, C++, Java, Python, SQL, JavaScript, HTML, CSS, JSON, SAS.
- **Developer Tools:** Terminal, Microsoft SQL Server, JupyterLab, XCode, VSCode, Anaconda Navigator, JupyterLab, Orange3.0, IBM SPSS, MS PowerBI, Advanced Excel, Git, SAS Studio.
- **Frameworks:** kinter, React, HDFS, Hive, Flask, Scikit-learn.

Certificates

- **Python for Everybody** | University of Michigan, *Coursera*.
- **Machine Learning Algorithms** | Supervised Learning Tip to Tail – Alberta Machine Learning Institute (AMII), *Coursera*.
- **Python Data Structures** | Rice University, *Coursera*.

Key Competencies

- **Good communication skills:** adept at conveying complex ideas both verbally and in writing.
- **Analytical thinking and problem-solving:** skilled at identifying issues, analyzing data, and proposing practical solutions to complex problems.
- **Adaptability and flexibility:** able to thrive in fast-paced, changing environments, and quickly adjust priorities to meet evolving needs.
- **Proficient in technical tools and software:** experienced in using Anaconda Navigator, jupyter notebook, MS office, Visual Studio code, JMP, and PowerBI, with a demonstrated ability to learn new technologies quickly.